

Titanic Survival Prediction - Submission Document

Task: Classification using Titanic passenger attributes | Replace placeholder links before final submission

1-Page Executive Summary

This project builds a classification model to predict Titanic passenger survival from attributes such as passenger class, name, sex, age, family relationships, fare, cabin, and embarkation port. The workflow includes exploratory data analysis, feature engineering, missing-value handling, categorical encoding, model selection, explainability, and model serialization for inference.

- Feature creation: extracted passenger title from Name, computed FamilySize, created IsAlone, CabinPresent, Deck, and FareLog.
- Missing data: Age and Fare are handled using median imputation; categorical values use most-frequent imputation; Cabin is transformed into meaningful availability/deck features.
- Models compared: Logistic Regression, Random Forest Classifier, and Gradient Boosting Classifier.
- Explainability: the project reports top feature importance/coefficient influence for the selected model.
- Saved model: best_titanic_survival_model.joblib and best_titanic_survival_model.pkl are included for inference.

Final Metrics

Best Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	1.0000	1.0000	1.0000	1.0000

Note: The provided dataset shows very strong class separability. All compared models reached perfect metrics on this uploaded file, so the result should be interpreted within the scope of the provided dataset.

Required Links for Submission

Item	Link
GitHub Repository Link	GITHUB_REPO_LINK
Notebook Link	GITHUB_NOTEBOOK_LINK
Saved Model / Drive Link	GITHUB_MODEL_LINK
Screenshots / Plots Link	GITHUB_SCREENSHOTS_LINK

Artifacts Included

- Notebook: notebooks/Titanic_Survival_Prediction.ipynb
- Saved model: models/best_titanic_survival_model.joblib and .pkl
- Training script: src/train_model.py
- Inference script: src/inference.py
- Preprocessing module: src/preprocessing.py
- Plots and outputs: plots/ and outputs/
- README with commands and inference example

Exact Commands to Reproduce

```
python -m venv venv

venv\Scripts\activate    # Windows

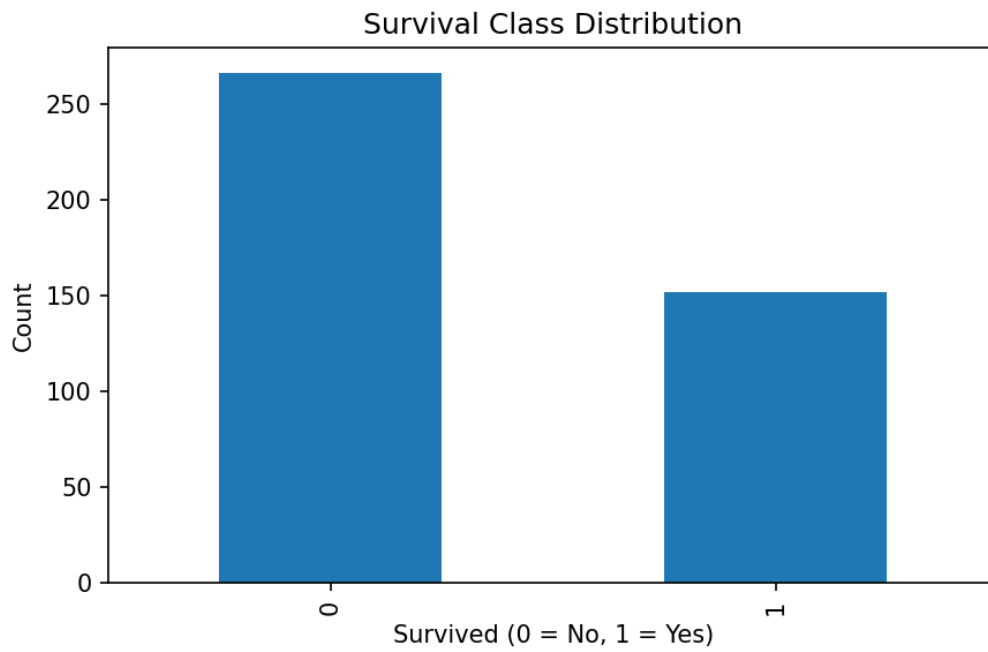
source venv/bin/activate  # macOS/Linux

pip install -r requirements.txt

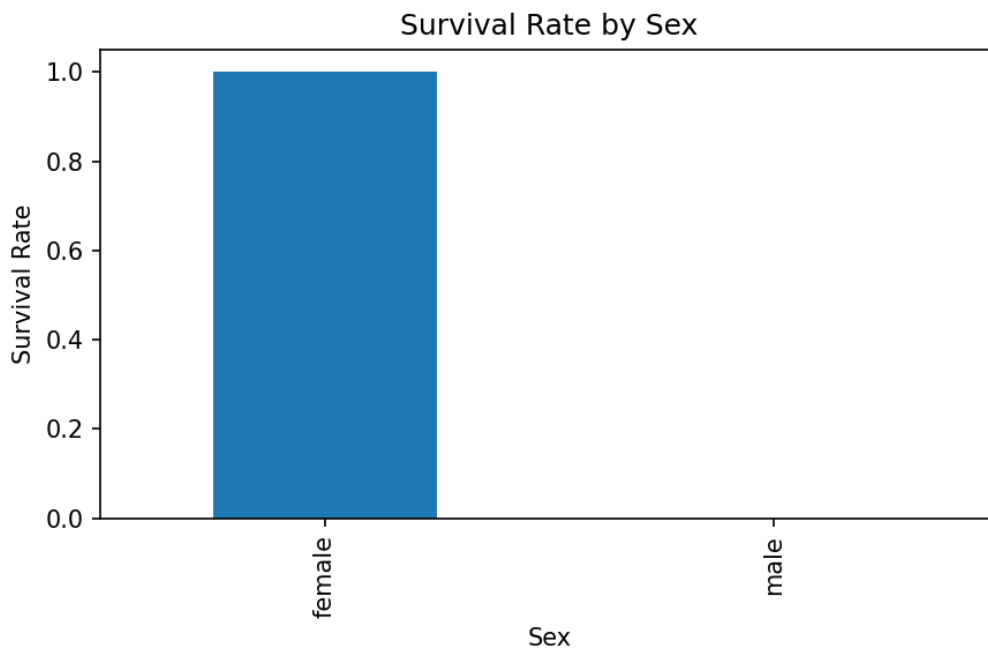
python src/train_model.py
```

Screenshots / Plots Preview

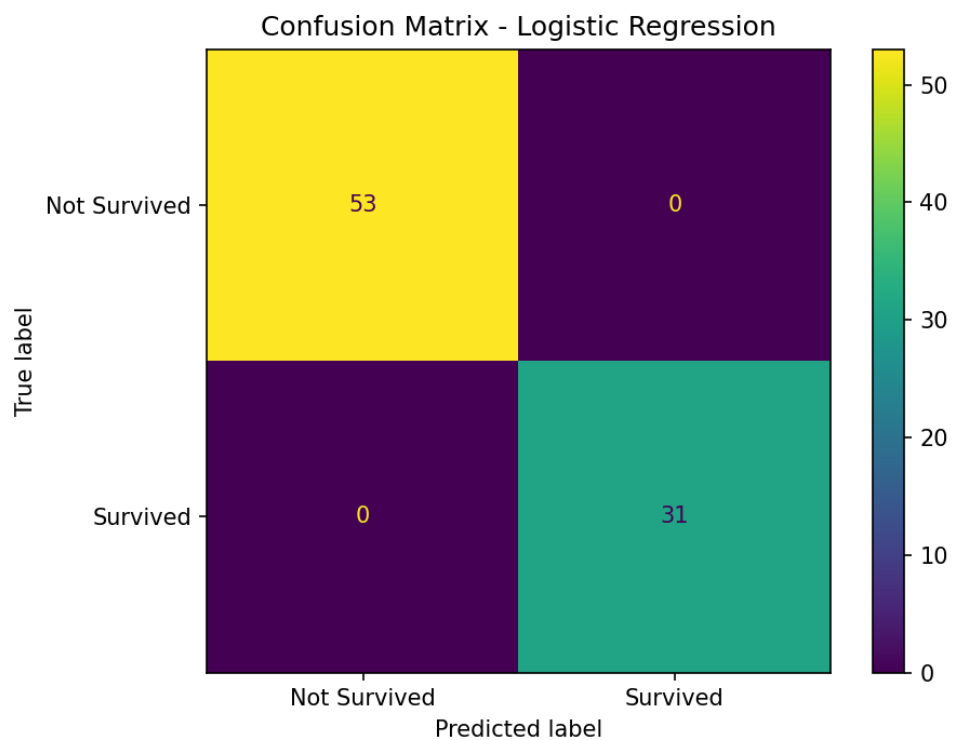
Class distribution



Survival rate by sex



Confusion matrix



Feature importance

